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Information Technology – Software Engineering

Project
AI EduAsistent – A Web Platform to Support Primary School Teachers

International Burch University

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1. Project Charter (with OKRs)

1.1 Business Case

Problem Statement:

There is an increasing need for creative, personalized, and high-quality teaching materials in modern education that align with contemporary pedagogical approaches. However, elementary school teachers often face challenges such as limited time for lesson preparation, a lack of resources, and a growing need for additional support in developing innovative teaching methods.
The integration of artificial intelligence into education represents both a new opportunity and a challenge, as many teachers still lack systematic support and practical understanding of how to incorporate these technologies into the teaching process responsibly.

Proposed Solution:
The AI EduAsistent project addresses these challenges by offering a simple web-based platform that enables teachers to generate teaching materials using the GPT-3 model automatically. By entering key parameters,subject, grade level, lesson unit, and type of material,teachers can quickly generate relevant and creative content tailored to the specific needs of their classrooms. In addition to content generation, the platform includes educational modules aimed at developing teachers' AI literacy by providing practical guidelines for the responsible and effective use of AI tools within an educational context.

Target Users:
The primary users are elementary school teachers who seek to improve the quality of their teaching materials and integrate modern technologies into their educational practices.

Key Benefits:

- **Time Efficiency:** Significant reduction in the time needed for lesson preparation.
- **Content Quality Improvement:** Enhanced creativity and relevance of teaching materials.
- **Development of AI Literacy:** Empowering teachers with foundational knowledge of artificial intelligence and its ethical application.
- **Accessibility:** A web-based platform that requires no installation and is accessible through any standard web browser.

Business Justification:
Given the current global emphasis on digital transformation in education, AI EduAsistent represents a timely and valuable solution. The platform not only supports teachers in enhancing the quality of their instructional materials but also contributes to the broader goal of integrating AI literacy into the education system. As such, AI EduAsistent positions itself as a potential partner for schools, educational institutions, and initiatives aimed at modernizing and enriching the teaching and learning process for both teachers and students.

1.2. SWOT Analysis

<table>
<tr><td rowspan="2">Internal</td><th>Strengths</th><th>Weaknesses</th></tr>
<tr><td>✓ Utilizes advanced GPT-3 technology for content generation
✓ Simple and intuitive interface designed specifically for teachers.
✓ No need for installation – accessible via web browsers.</td><td>✓ Dependence on the stability and accessibility of the OpenAI API.
✓ Requires a continuous internet connection for full functionality.
✓ Potential costs associated with API usage for larger user bases.
✓ Risk of potential misuse or misunderstanding of AI-generated content.</td></tr>
<tr><td rowspan="2">External</td><th>Opportunities</th><th>Threats</th></tr>
<tr><td>✓ Growing interest in the use of AI in education.
✓ Opportunities for partnerships with schools and educational institutions.
✓ Potential to expand to higher educational levels (secondary, university).</td><td>✓ Possible changes in OpenAI API usage policies.
✓ Emerging competitors offering similar AI-powered educational tools.
✓ Limited digital literacy among some segments of the teaching population.</td></tr>
</table>

1.3. Project Structure and Implementation Plan

1.3.1. Vision and Objectives (OKRs)

Vision:
To create an accessible and user-friendly web platform that empowers elementary school teachers to efficiently generate high-quality and creative teaching materials using artificial intelligence, while simultaneously developing their AI literacy through practical and responsible application.

Objectives and Key Results (OKRs):

Objective 1: Develop a functional application for the automatic generation of teaching materials.

- KR1: Enable content generation for four types of materials: creative assignment, homework, educational game, and project-based assignment.
- KR2: Integrate the OpenAI GPT-3 API and complete testing by July 2025.

Objective 2: Increase AI literacy among teachers through educational content.

- KR1: Develop and publish three educational pages covering the basics of AI, prompt creation, and ethical guidelines.
- KR2: Achieve at least 50 views of the educational pages during the pilot phase.

1.3.2. Project Description

AI EduAsistent is a web-based platform designed to support elementary school teachers in creating creative teaching materials tailored to specific subjects, grade levels, and lesson units. By entering basic parameters, teachers can automatically generate creative and relevant content.
In addition, the platform offers educational modules aimed at helping teachers responsibly and effectively integrate AI technologies into the educational process.

1.3.3. Main Functionalities

- Input form for subject, grade, lesson unit, and type of material.
- Automatic content generation using the GPT-3 model via the OpenAI API.
- Detection and reuse of previously generated materials to optimize API calls.
- Clear display of generated results with the option to copy the content for classroom use.
- Educational modules covering AI fundamentals, prompt crafting, and ethical use of AI tools.

1.3.4. Stakeholders

Successful project management requires a thorough identification and analysis of stakeholders to ensure timely recognition of their needs, expectations, and potential impact on the project flow. The **AI EduAsistent** project involves the following key stakeholder groups:

Primary School Teachers:
The main users of the application, whose needs for simple, fast, and creative generation of teaching materials directly shape the system's functional and technical requirements. Their feedback will be crucial for the continuous improvement of the application.

Students:
Although not direct users of the application, students are the ultimate beneficiaries of the educational materials created through the platform. Their level of engagement, motivation, and educational achievements serve as an indirect but essential measure of the project's success.

Educational Institutions:
Responsible for approving the implementation of new technologies in educational processes, educational institutions are interested in tools that enhance the quality of teaching while complying with regulatory standards and educational policies.

Software Development Team:
This group includes developers, designers, and QA engineers responsible for the implementation, testing, and maintenance of the application. Their collaboration and technical expertise are critical for the timely delivery of high-quality functionality.

OpenAI (AI Technology Provider):
As the provider of the GPT model through API integration, OpenAI has a direct influence on the application's performance, reliability, and operational costs. Changes in API policies, pricing models, or technical specifications may potentially affect the sustainability of the project.

Effective communication with all identified stakeholders and timely incorporation of their feedback into the development process will be one of the key factors for the project's success.

1.3.5. Technical Architecture

The technical architecture of the AI EduAsistent project defines the core structure and technologies used for the development, integration, and maintenance of the application. The system is designed as a three-tier architecture, enabling modularity, scalability, and ease of maintenance.

Frontend (User Interface):

- Technologies: HTML5, CSS3, JavaScript
- Description:
  The frontend enables users (teachers) to intuitively input data (teaching unit, subject, grade level, type of material) through simple and well-organized forms.
  The focus is on accessibility, responsive design, and delivering a seamless user experience.

Backend:

Technologies: PHP, REST API

- Description:
  The backend processes user requests, communicates with the database, and interacts with the OpenAI API. It implements the logic for handling user input and sends prompts to the GPT-3 model to generate personalized teaching materials.

Database:

- Technologies: MySQL
- Description:
  The database stores user inputs, the history of generated materials, and user information (in compliance with data protection standards).
  The database structure is optimized for fast data retrieval and easy scalability.

AI Integration:

- Technologies: OpenAI GPT-3 API
- Description:
  The application connects to the OpenAI GPT-3 model via an API key, sending customized prompts based on user inputs.
  The GPT-3 model's responses are processed and formatted for display to the user.

Security and Data Protection:

- HTTPS protocol is used for secure data transmission.
- API keys and access tokens are securely stored on the server side.
- Basic security practices are implemented, including input validation, protection against SQL injection attacks, and basic user authentication.

1.3.6. Methodology

For the implementation of the AI EduAsistent project, an agile software development methodology was selected, with a particular reliance on the Scrum framework. This approach enables iterative and incremental product development, which is crucial in the context of evolving user requirements and the rapid advancement of artificial intelligence technologies.

Key characteristics of the selected methodology include:

Iterative development:
The project was divided into short development cycles (sprints) lasting two to three weeks, with clearly defined goals for each cycle.

Continuous testing and evaluation:
At the end of each sprint, an evaluation of the developed functionalities was conducted from the

perspective of the end-user, whereby the application creator, as an active teacher, assessed the user experience, intuitiveness, and relevance of the generated content. This reflective analysis enabled the timely identification of necessary improvements and enhancements to the application.

Flexibility:
Changes in requirements and functionalities could be quickly integrated thanks to adaptive planning.

Transparency:
Regular Scrum meetings (daily stand-ups) and sprint retrospectives ensured process transparency and supported continuous improvement of team efficiency.

User-centered approach:
Special emphasis was placed on understanding user needs by creating user stories and defining acceptance criteria.

The agile methodology enabled rapid adaptation to new insights in educational technology and ensured a high degree of alignment between the application's functionalities and teachers' real needs.

1.3.7. Timeline

The AI EduAsistent project is planned as a series of six sprints, each lasting approximately two weeks. This organization enables rapid functionality development, evaluation, and adjustment to user requirements.

- **Sprint 1: Requirements Analysis and Initial Planning**
  - Defining the key functionalities and system architecture
  - Preparing user stories and initial designs
- **Sprint 2: System Foundation Setup**
  - Creating the frontend structure (data input forms)
  - Implementing basic backend processes (PHP + API integration)
- **Sprint 3: Database and Core Functionalities**
  - Designing and connecting the MySQL database
  - Developing the core logic for communication with the OpenAI API
- **Sprint 4: Development of Educational Modules**
  - Creating educational pages for teachers (AI technologies, ethical guidelines)
  - Improving the user interface and overall design
- **Sprint 5: Testing and Security Optimization**
  - Internal testing of all functionalities
  - Implementing basic security measures (input validation, database protection)
- **Sprint 6: Finalization and Deployment Preparation**
  - Final performance optimization
  - Writing user manuals and technical documentation
  - Preparing the platform for production deployment

This plan enables efficient development and testing of the application, ensuring that both main components, content generation and educational modules, are completed within the projected timeframe.

1.3.8. Risks and Constraints

During the development and implementation of the AI EduAsistent project, several potential risks and constraints were identified that could impact the project's success:

Technical Risks:

- **Dependency on External Services:**
  The platform relies on the stability and accessibility of the OpenAI GPT-3 API. Any changes in API usage policies, service availability, or pricing could disrupt the functionality or increase operational costs.
- **Security Vulnerabilities:**
  As a web-based application that handles user data, the platform must implement strong security measures (e.g., HTTPS, data validation) to prevent data breaches and ensure user trust.

Operational Risks:

- **Limited User Feedback:**
  Due to the dual role of the developer as both creator and initial user (a practicing teacher), the evaluation may lack broader external feedback, potentially delaying the identification of usability issues.
- **Limited Digital Literacy among Users:**
  Some segments of the teaching population may have limited experience with digital tools, which could slow down adoption and effective use of the platform.
- **Scalability Challenges:**
  Significant growth in the number of users beyond initial estimates could require optimization and scaling of the server infrastructure.

Resource Constraints:

- **Financial Limitations:**
  With a modest initial budget, the project might face limitations in acquiring advanced development tools, paid services, or additional technical resources.

Legal and Ethical Constraints:

- **Compliance with Data Protection Regulations:**
  The application must comply with relevant data protection laws (e.g., GDPR) when collecting and processing user information.
- **Responsible Use of AI-Generated Content:**
  Ensuring that AI-generated content is appropriate, unbiased, and educationally relevant is a continuing ethical obligation.

Mitigation Strategies:

- Developing contingency plans for service interruptions.
- Building a modular and scalable system architecture.
- Conducting regular security audits and improvements.
- Gradually expanding the platform based on user demand and available resources.
- Providing clear educational materials to support teachers with varying levels of digital literacy.

By proactively identifying and managing these risks, the project aims to minimize potential disruptions and ensure the delivery of a reliable, responsible, and user-friendly educational platform.

1.3.9. Success Measurement

The success of the **AI EduAsistent** project will be evaluated based on the achievement of key performance indicators (KPIs) that are aligned with the project's objectives. Both qualitative and quantitative measures will be utilized to ensure a comprehensive assessment of the platform's efficiency and impact.

Key Success Indicators:

Functionality Achievement:
Successful implementation of all core functionalities, including intuitive data input forms, reliable integration with the OpenAI API, secure database storage, and safe processing of user data.

User Satisfaction:
Positive feedback from teachers regarding the ease of use, relevance, and creativity of the generated teaching materials, collected through surveys and informal evaluations.

Operational Stability:
Consistent platform performance without major technical issues during internal testing phases and initial deployment.

Educational Impact:
Demonstrated improvement in the creativity of teaching materials, contributing to greater student engagement and enhanced learning outcomes, based on teacher observations.

Security and Compliance:
Full compliance with data protection standards (e.g., GDPR) and successful implementation of necessary security measures (SSL, data validation, access control).

Adoption and Usage Rates:
Monitoring the number of teachers actively using the platform and analyzing engagement levels over time.

Scalability Readiness:
The platform's ability to support an increasing number of users without significant degradation in performance.

Evaluation Approach:

- Internal testing reports and quality assurance documentation
- User surveys and feedback forms
- System performance monitoring and API usage statistics

By systematically measuring these indicators, the project aims to ensure not only the technical success of the platform but also its educational value and long-term sustainability.

1.3.10. Assumptions

The AI EduAsistent project is built on several key assumptions that form the basis for development planning and risk assessment:

- Stable access to the OpenAI GPT-3 API will be available throughout development and deployment.
- Elementary school teachers possess the basic digital **literacy** required to use the web platform without extensive training.
- Internet connectivity is reliably available in school environments where the platform will be used.
- No major shifts in educational policies will occur that would restrict the use of AI tools in teaching.
- User feedback will be provided during testing and will reflect realistic teaching scenarios.

These assumptions will be monitored continuously and updated as needed.

1.3.11. Approval

This document represents the official project charter for AI EduAsistent. It has been reviewed and approved by the project owner and supervising authority.

Approved by:
Jasmina Hasanović – Project Lead
Jasmina Hasanović – Supervisor
Date: 12.05.2025

Visual Project Overview

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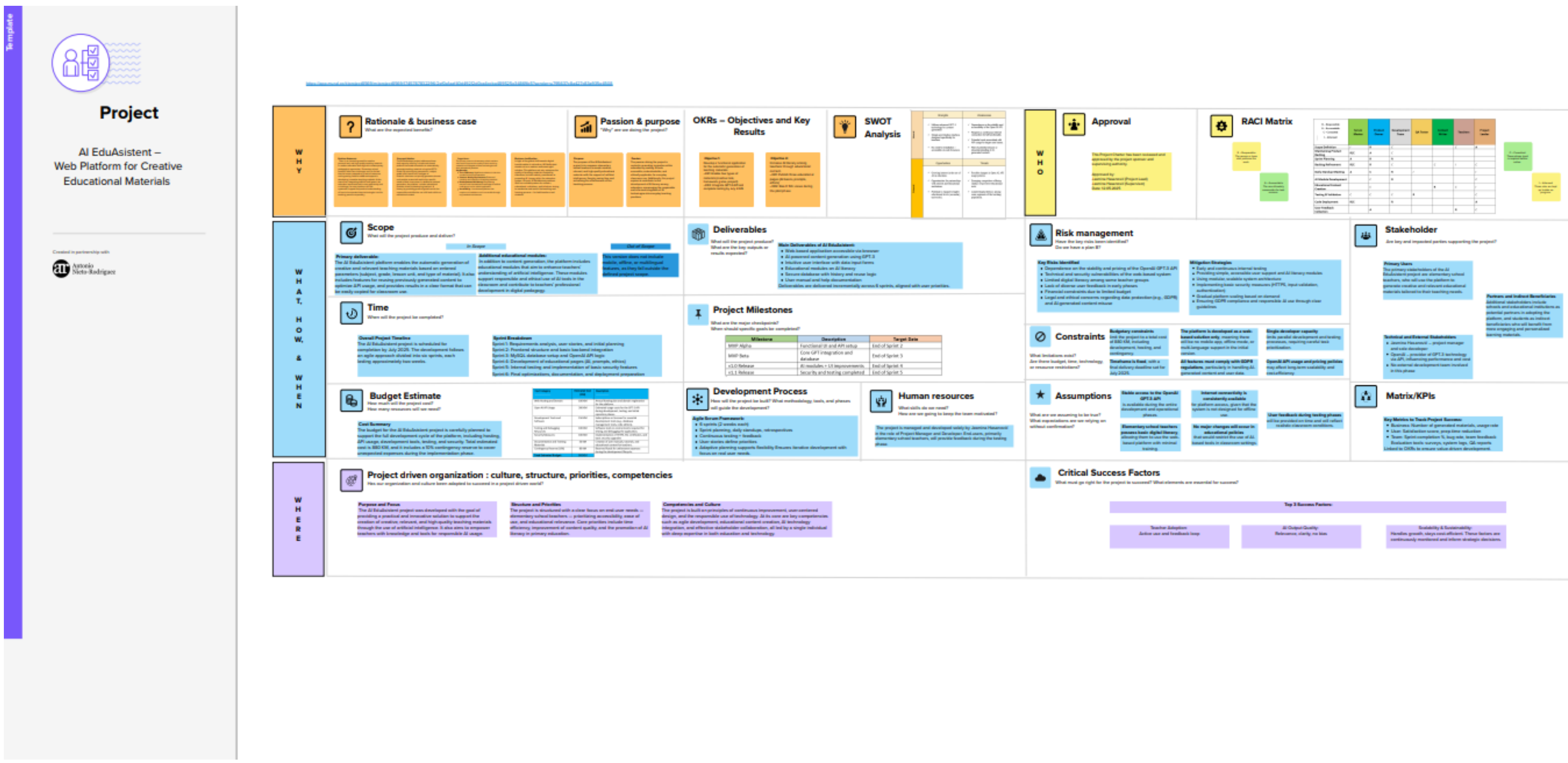

2. Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) represents a hierarchical decomposition of the AI EduAsistent project into manageable sections of work. It helps define all deliverables and activities required to complete the project, ensuring clarity, task ownership, and better planning throughout the development lifecycle. The WBS is aligned with the Agile methodology and mirrors the structure of the planned sprints and functionalities.

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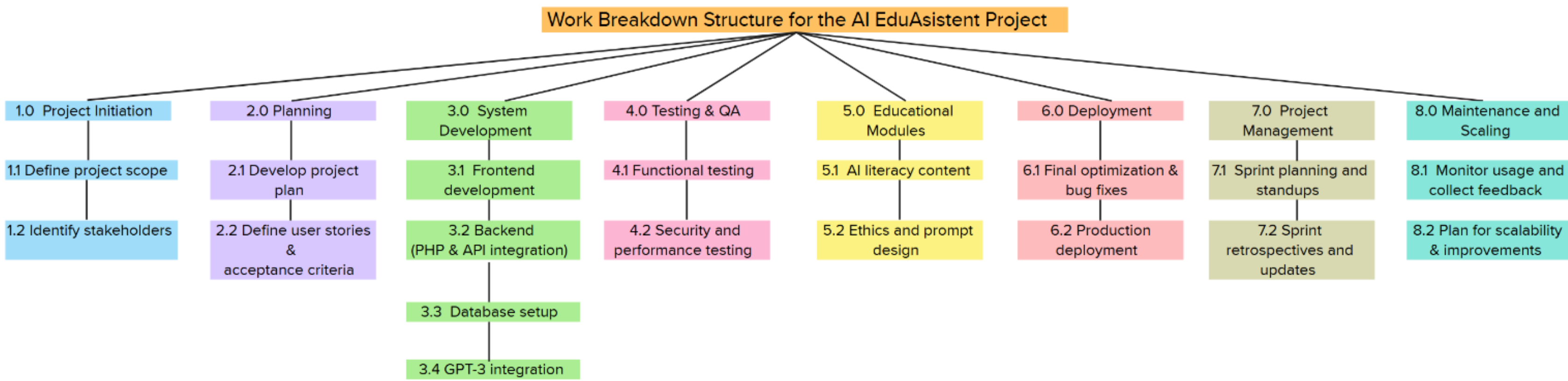

Figure 1: Visual Work Breakdown Structure for AI EduAsistent Project

3. Product Roadmap and Timeline

The Product Roadmap outlines the development timeline of the AI EduAsistent platform across six sprints, aligned with Agile methodology. Each sprint includes clearly defined goals, deliverables, and a version milestone. The roadmap reflects an incremental delivery approach, enabling continuous feedback and adaptation while ensuring the timely release of core functionalities and supporting modules.

	Sprint	Duration	Goals	Deliverables / Milestones	Sprint Title
1	Sprint 1	2 weeks	Requirements analysis, system architecture, and initial designs	User stories, architecture diagram, sprint plan	Initial Planning
2	Sprint 2	2 weeks	Frontend structure, basic backend processes, API setup	Functional frontend forms, API connection working	Core Functionality Setup
3	Sprint 3	2 weeks	Database design, OpenAI API integration core logic	Database operational, content generation feature ready	Backend & AI Integration
4	Sprint 4	2 weeks	Develop educational modules and UI improvements	Published AI literacy content, improved UX/UI	Educational Content Development
5	Sprint 5	2 weeks	Testing functionalities, apply security improvements	Test reports, security validation	Testing & Security
6	Sprint 6	2 weeks	Final optimization, documentation, production deployment	Deployment-ready platform, getting started guide	Finalization and Deployment

4. Product Scope and Requirement Prioritization

4.1 Overview

The product scope of the AI EduAsistent platform consists of key functional modules that allow users,elementary school teachers,to generate tailored teaching materials using GPT-3. In addition, educational support modules raise awareness of artificial intelligence and its ethical application in education. This section provides a structured breakdown of the platform's features and applies a prioritization technique to plan the development process.

4.2 Functional Breakdown

Core Functionalities

- Input form for subject, grade level, lesson unit, and type of material
- GPT-3 powered automatic content generation
- Display and copy function for the generated results
- Database for storing and reusing previously generated content

Educational Features

- Informational page about AI fundamentals
- Guidelines for writing effective prompts
- Ethical usage instructions for AI in the classroom

Supporting Features

- Basic input validation and protection (e.g., SQL injection prevention)
- Backend support for scalability and performance monitoring
- Optional admin-level analytics (for future use)

4.3 Prioritization Technique: MoSCoW

To prioritize features and functionalities for incremental development, the **MoSCoW method** is used. This framework categorizes each feature as:

- **Must have** – critical for project success
- **Should have** – important but not essential for MVP
- **Could have** – nice to include if time/resources allow
- **Won't have for now** – not planned for the current development cycle

4.4 Prioritization Table

Feature	Category	Priority
Input form: subject, grade, lesson unit, material type	Core	Must
GPT-3 content generation	Core	Must
Display and copy of results	Core	Must
Storage and retrieval of previous outputs	Core	Should
AI basics educational page	Educational	Must
Prompt creation guidelines	Educational	Should
Ethical AI use education	Educational	Should
Input validation & SQL injection protection	Supporting	Must
Usage statistics and performance monitoring	Supporting	Could
Admin-level analytics dashboard	Supporting	Won't

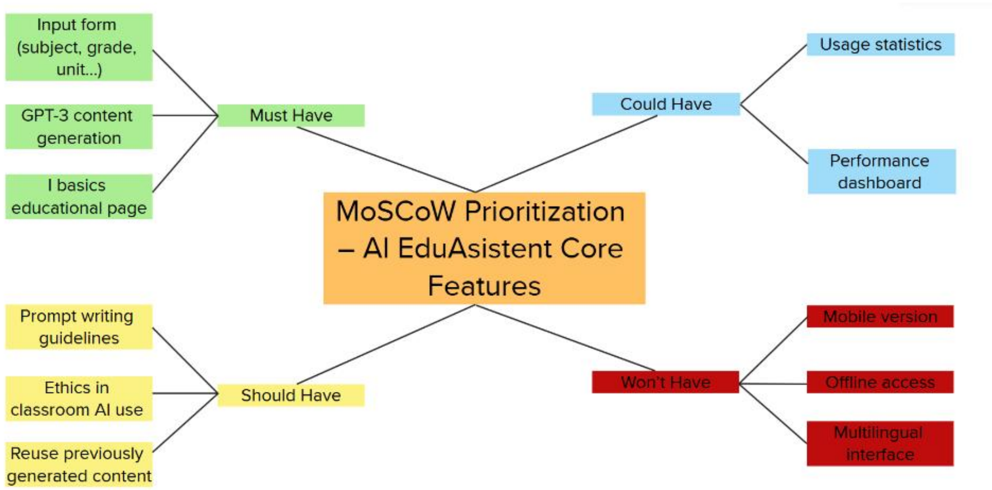

Figure 2: MoSCoW Prioritization Visualization for AI EduAsistent Functionalities

5. User Requirements Specification

5.1 Overview

The User Requirements Specification (URS) defines both functional and non-functional needs of the users based on system goals and stakeholder expectations. These requirements serve as the foundation for system design, development, and testing. The goal is to ensure clarity, traceability, and alignment between user needs and technical implementation.

5.2 URS Template

Section	Description
Requirement ID	Unique identifier (e.g., FR-001, NFR-001)
Requirement Type	Functional / Non-Functional
Requirement Description	Clear and concise explanation of the user's need
Priority	Must / Should / Could / Won't
Acceptance Criteria	The objective conditions under which the requirement is considered fulfilled
Source / Stakeholder	Who identified or needs this requirement (e.g., teacher, dev team)

5.3 Functional Requirement Example

Section	Description
Requirement ID	FR-001
Requirement Type	Functional
Requirement Description	The system shall allow the user to input the subject, grade level, lesson unit, and material type through a single form.
Priority	Must
Acceptance Criteria	The form includes all four fields and can be submitted successfully. All fields must be validated for input format.
Source / Stakeholder	Primary School Teacher

5.4 Non-Functional Requirement Example

Section	Description
Requirement ID	NFR-001
Requirement Type	Non-Functional
Requirement Description	The system shall ensure that all communication with the OpenAI API is conducted over HTTPS for secure data transmission.
Priority	Must
Acceptance Criteria	All API requests are encrypted and no unsecured protocols are used. Traffic is inspected during testing.
Source / Stakeholder	Development Team / Legal Compliance

6. Resource Breakdown Structure (RBS)

6.1 Overview

The Resource Breakdown Structure (RBS) provides a hierarchical categorization of all the resources required for the AI EduAsistent project. It includes human resources, technical tools, software platforms, and infrastructure elements. This breakdown supports cost estimation, scheduling, and resource allocation throughout the project lifecycle.

6.2 RBS Categories and Elements

Category	Resource Name / Type	Usage
Human Resources	Project Manager	Overall coordination and project oversight
	Backend Developer (PHP/API)	System logic and integration with OpenAI
	Frontend Developer	UI/UX design and implementation
	QA Tester	Functional and security testing
	Content Designer (AI Literacy)	Writing educational pages on AI, ethics, and prompt use
	Technical Writer	User manuals and documentation
Software Tools	Visual Studio Code, phpMyAdmin	Development and database management
	Miro / Mural	Brainstorming and visual planning
	Trello / Jira	Agile project tracking and sprint planning
AI Resources	OpenAI GPT-3 API	Content generation based on teacher input
Infrastructure	Web Hosting & Domain	Platform hosting and deployment
	MySQL Database	Storage of generated content and user input
	SSL Certificate / HTTPS Protocol	Secure communication and data privacy
Documentation	Microsoft Word / Google Docs	Report writing, planning documentation
	GitHub	Internal technical documentation

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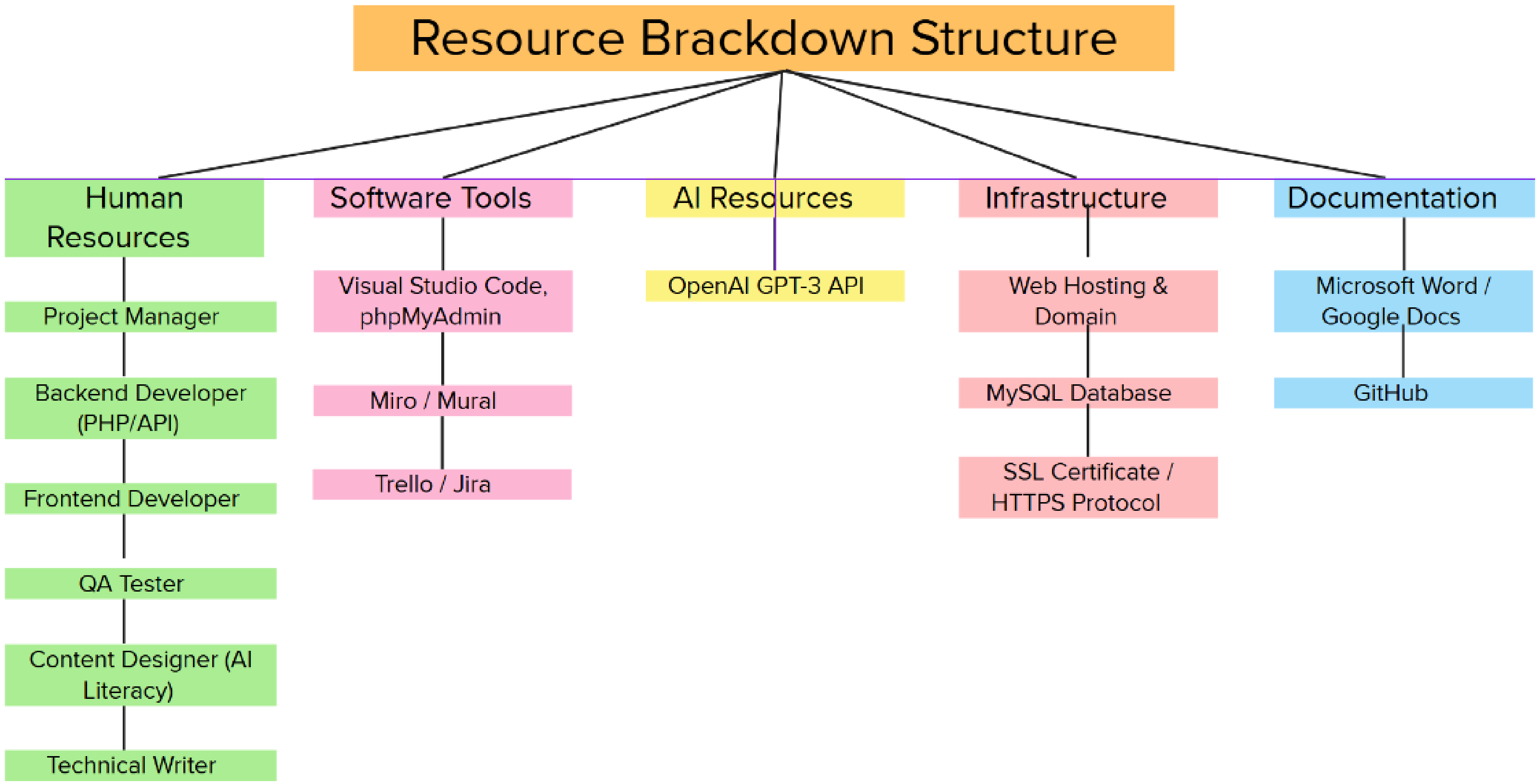

Figure 3: Visual Resource Breakdown Structure (RBS) for the AI EduAsistent Project

7. Stakeholder Map

7.1 Overview

The stakeholder map visualizes the individuals and organizations who influence or are impacted by the AI EduAsistent project. Stakeholders are categorized by their level of power and interest, enabling the project team to prioritize communication, involvement, and feedback based on their position.

7.2 Stakeholder Matrix

Stakeholder	Type	Power	Interest	Category
Primary School Teachers	Primary	High	High	Manage Closely
Students	Indirect/Secondary	Low	High	Keep Informed
Educational Institutions	Organizational	High	Medium	Keep Satisfied
Software Development Team	Internal	Medium	High	Manage Closely
OpenAI (API Provider)	External Partner	High	Low	Monitor

Categories are based on the classic Power-Interest Matrix:

- **Manage Closely** = High power, high interest
- **Keep Satisfied** = High power, low interest
- **Keep Informed** = Low power, high interest
- **Monitor** = Low power, low interest

7.3 Visual Stakeholder Map

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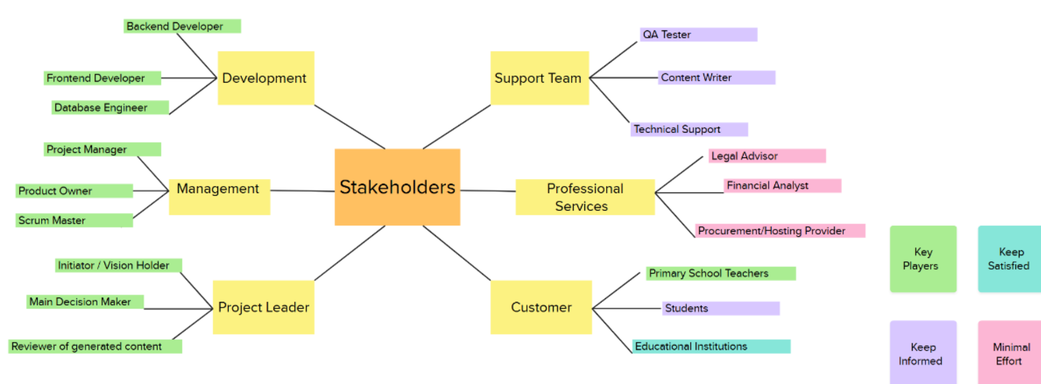

Figure 4: Power–Interest Stakeholder Map for AI EduAsistent

8. RACI Matrix

8.1 Overview

The RACI Matrix (Responsible, Accountable, Consulted, Informed) outlines the distribution of roles and responsibilities among key stakeholders in the AI EduAsistent project. It ensures that each activity within the project is assigned, helping to avoid confusion, clarify accountability, and improve collaboration between team members.

- **R – Responsible**: The person(s) who actually perform the task or activity
- **A – Accountable**: The person who is ultimately answerable for the correct and thorough completion of the task
- **C – Consulted**: The person(s) whose opinions are sought; Typically subject matter experts
- **I – Informed**: The person(s) who need to be kept informed of progress and results

This RACI matrix reflects the agile methodology and collaborative nature of the project team, ensuring that internal and external stakeholders are properly engaged.

8.2 RACI Matrix Table

Activity	Project Leader	Product Owner	Dev Team	QA Tester	Content Writer	Teachers	Institutions
Scope Definition	A	R	C			C	I
Sprint Planning	A	R	R				
Backlog Refinement	C	R	C		C	I	
Daily Standups	I	A	R	I			
AI Module Development (GPT Integration)	C	C	R				
Educational Content Creation	I	C			R	C	I
Testing and Validation	C	C	C	R			I
User Feedback Collection	C	A				R	C
Deployment and Hosting Setup	A	R	R				
Compliance and Data Privacy Review	C	A					R

Figure 5a: Primary RACI Matrix for Key Activities in AI EduAsistent Project

R – Responsible: **A – Accountable:** **C –Consulted:** **I – Informed:**	**Scrum Master**	**Product Owner**	**Development Team**	**QA Tester**	**Content Writer**	**Teachers**	**Project Leader**
Scope Definition	*I*	**R**	*C*			*C*	**A**
Maintaining Product Backlog	**R/C**	**A**	*C*				
Sprint Planning	**A**	**R**	**R**				
Backlog Refinement	**R/C**	**R**	*C*		*C*	*I*	*C*
Daily Standup Meeting	**A**	**C**	**R**	*I*			*I*
AI Module Development		*C*	**R**				*C*
Educational Content Creation		*C*			**R**	*C*	*I*
Testing & Validation	*C*	*C*	*C*	**R**			*C*
Code Deployment	**R/C**		**R**				**A**
User Feedback Collection		**A**				**R**	*C*

Figure 5b: RACI Matrix Based on Scrum Roles

Activity	Project Leader (Jasmina)	Product Owner	Developer	QA Tester	Content Writer	Teachers	Institutions
Planning / Schedule	R	A	C			I	I
Risk Management	A	C				I	R
Quality Management	A	C	R	C	C	I	
Procurement	A						R
1. Specifications Listing	R	C	R		C	C	
2. Site Requirements	R	C	R			C	R
3. Call for Tenders *(optional)*	A						R
4. Budget Approval *(optional)*	A						R
5. Contract Negotiations *(optional)*	A						R
Educational Content Creation	A	C			R	C	I

R – Responsible **A** – Accountable **C** – Consulted **I** – Informed

Figure 6c: RACI Matrix for Roles and Responsibilities in AI EduAsistent Project

9. Communication Plan – Channels, Frequency, Purpose

The communication plan ensures that all project participants are informed, aligned, and actively engaged throughout the project lifecycle. The following table outlines the main communication channels, their frequency, and specific purposes per stakeholder group.

Stakeholder	Channel	Frequency	Purpose
Project Lead	Email, in person	Weekly	Progress updates
Teachers (Pilot Group)	Surveys, Feedback	End of Sprint	Usability feedback
Mentor / Supervisor	Email, Report	Bi-weekly	Oversight, academic review
Self (as Dev)	Trello / Daily Log	Daily	Task management
OpenAI API	Dashboard, email	As needed	Technical integration monitoring

10. Metrics & KPIs

To track progress, evaluate value delivery, and ensure team efficiency throughout the development of the AI EduAsistent platform, I defined a clear set of metrics. These indicators are aligned with both our project OKRs and the agile delivery timeline, allowing for systematic monitoring and timely adjustments. The metrics fall into three main categories: business value, customer value, and team effectiveness.

Business Value Metrics

These metrics reflect how the project contributes to educational institutions' broader goals, such as digitalization, cost-efficiency, and pedagogical innovation.

Metric	Description	Target	Related to
Adoption Rate (Pilot Schools)	Number of schools actively using the platform during the pilot phase	5 schools	OKR 1 / KR2
Time Saved per Teacher	Average weekly time saved in lesson preparation	60 minutes	Business Case: Time Efficiency
Cost Savings on Printed Materials	Reduction in school expenses for printed educational content	20%	Business Justification
Volume of Generated Content	Monthly number of AI-generated lesson materials	200+	OKR 1 / Functionality Goal

Customer Value Metrics

These indicators focus on the experience and satisfaction of the teachers using the platform, as well as their engagement with AI literacy content.

Metric	Description	Target	Related to
Sprint Goal Achievement Rate	Completion rate of planned sprint objectives	100% per sprint	Agile Sprint Review
Velocity (Story Points)	Average number of story points delivered per sprint	≥ 25	Scrum Tracking
Defect Density	Number of bugs per 1,000 lines of code	≤ 2	QA – Sprint 5
Time to Resolve Critical Bugs	Time from bug identification to resolution for high-priority issues	< 48 hours	Dev & QA Process

Connection to OKRs and Roadmap

These metrics are directly tied to our project's core objectives:

- Objective 1 is supported by adoption, usage, and satisfaction metrics.
- Objective 2 is monitored through engagement with educational content.
- Sprint 5 and 6 focus on validating quality, gathering feedback, and measuring success prior to full release.
- Agile metrics help maintain team discipline, transparency, and continuous improvement across all sprints.

By consistently tracking and evaluating these indicators, I ensure that the project stays aligned with its value-driven goals, adapts to user needs, and delivers a meaningful impact in the classroom.

11. Risk Plan

To ensure the stability and sustainability of the AI EduAsistent project, I have identified key risks that could potentially affect development, deployment, or user adoption. Each risk has been assessed in terms of its likelihood and potential impact. I have also outlined mitigation strategies, triggers, and predefined actions in case the risk materializes.

11.1 Identified Project Risks and Mitigation Strategie.

Risk	Likelihood	Impact	Trigger	Mitigation Strategy	Action if Realized
OpenAI API becomes	Medium	High	API downtime or pricing policy change	Monitor API usage; set alerts;	Pause release; use contingency

unavailable or restricted				consider backup AI	funds; explore alternatives
Low teacher feedback during testing phase	High	Medium	Fewer than 30% of teachers respond	Provide guided forms and small incentives	Conduct 1-on-1 interviews; extend testing period
Low adoption due to limited digital literacy	Medium	Medium	Low login and usage rates	Provide onboarding videos and manuals	Offer live onboarding sessions; collect insights for simplification
Budget overrun due to high API usage	Medium	High	Monthly costs exceed forecast	Set usage caps; monitor billing dashboard	Limit functionality temporarily; activate contingency reserve
Institutional concerns about data privacy	Low	High	Formal complaints or lack of approval	Ensure GDPR compliance; communicate clearly	Publish privacy policy and security documentation; host meetings
Curriculum policy changes affect relevance	Low	Medium	Updated national teaching standards	Design prompts and modules to be modular and flexible	Revise prompt suggestions; update content types
Team availability issues	Medium	Medium	Sprint delays due to absences or overload	Use Trello for workload tracking; document development procedures	Reassign tasks; extend sprint if necessary

11.2 Risk Monitoring and Control

All risks will be monitored continuously through:

- Weekly sprint reviews and retrospectives
- API usage dashboards
- Direct teacher feedback channels
- Trello task status and velocity tracking
- Budget monitoring dashboards

Trigger indicators are pre-defined for each risk, and actions are clearly outlined to reduce response time.

11.3 Visual Risk Display

To complement the detailed risk register and support agile planning, key project risks have been visually represented on the Mural board using color-coded sticky notes. Each note includes the risk title, likelihood, impact, mitigation strategy, and predefined action. The color scheme reflects severity and type:

- **Red** – High likelihood and high impact (critical risk)
- **Orange –** Medium likelihood or high impact (moderate risk)
- **Yellow** – Low likelihood and/or medium impact (low risk)
- **Purple –** Technical or cost-related risks with structural implications

This format provides a clear, prioritized overview of potential threats, supporting quick assessment and informed sprint planning.

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12. Budget Control Plan

The Budget Control Plan ensures that all financial resources allocated to the AI EduAsistent project are used efficiently, transparently, and aligned with project goals. This section outlines the approved budget, the mechanisms for monitoring and control, key warning signals (red flags), and concrete actions to take in the event of deviations.

12.1 Approved Budget Summary

Cost Category	Estimated Cost (KM)	Description
Web Hosting and Domain	100 KM	Annual hosting plan and domain registration for the platform.

Open AI API Usage	300 KM	Estimated usage costs for the GPT-3 API during development, testing, and initial operation phases.
Development Tools and Software	150 KM	Subscriptions or licenses for essential development tools (e.g., database management tools, code editors).
Testing and Debugging Resources	100 KM	Software tools or environments required for testing and debugging the application.
Security Measures	100 KM	Implementation of HTTPS, SSL certificates, and basic security upgrades
Documentation and Training Materials	50 KM	Creation of user manuals, tutorials, and educational content for teachers.
Contingency Reserve (10%)	80 KM	Reserved funds for unforeseen expenses during the development lifecycle.
Total Estimated Budget:	880KM	

12.2 Budget Monitoring and Control Mechanisms

To prevent cost overruns and ensure optimal use of resources, the following control processes are in place:

- **Sprint-Based Budget Review:** At the end of each sprint (every 2 weeks), current spending is reviewed against the allocated budget per cost category.
- **Monthly Cost Reporting:** A simple expense tracking sheet is maintained, recording actual vs. planned spending.
- **Approval for Additional Costs:** Any non-budgeted expenses must be approved by the project lead and recorded with justification.
- **Contingency Activation Threshold:** If 90% of the total budget is used before final deployment, contingency funds are unlocked for controlled use.

12.3 Budget Red Flags (Triggers for Action)

Red Flag	Trigger Threshold	Implication	Immediate Action
OpenAI API costs exceed 250 KM	>85% of planned usage	Potential risk of overrun	Optimize prompt design, reduce test volume, activate contingency
Hosting/SSL unexpectedly increases	Any renewal or upgrade not budgeted	Unplanned fixed cost	Negotiate with provider or downgrade hosting plan
Dev tools exceed 150 KM	Subscriptions go beyond plan	Non-critical overhead	Switch to open-source tools, pause upgrades
Total budget exceeds 880 KM	Full overrun	Critical financial breach	Freeze feature expansion, postpone low-priority tasks

12.4 Budget Response Strategy

If a red flag is triggered, the following actions will be taken:

- **Review and re-prioritize scope** using MoSCoW model (postpone "Could have" features)
- **Activate contingency funds** only for features marked as "Must have"
- **Temporarily halt new expenses** until current spending is analyzed
- **Document and justify** all financial deviations and decisions

12.5 Roles and Responsibilities in Budget Control

Role	Responsibility
Project Lead	Overall budget ownership; approval of expenses and contingency activation
Development Team	Reporting technical tool/API needs per sprint
QA Tester	Communicating need for paid testing tools or security upgrades
Supervisor / Mentor	Verifying alignment with academic expectations and providing budget feedback during reviews

This structured approach to budget monitoring and response ensures the financial health of the AI EduAsistent project throughout its lifecycle. The team can act swiftly and maintain control even in unexpected situations by combining real-time tracking with clearly defined thresholds and responsibilities.

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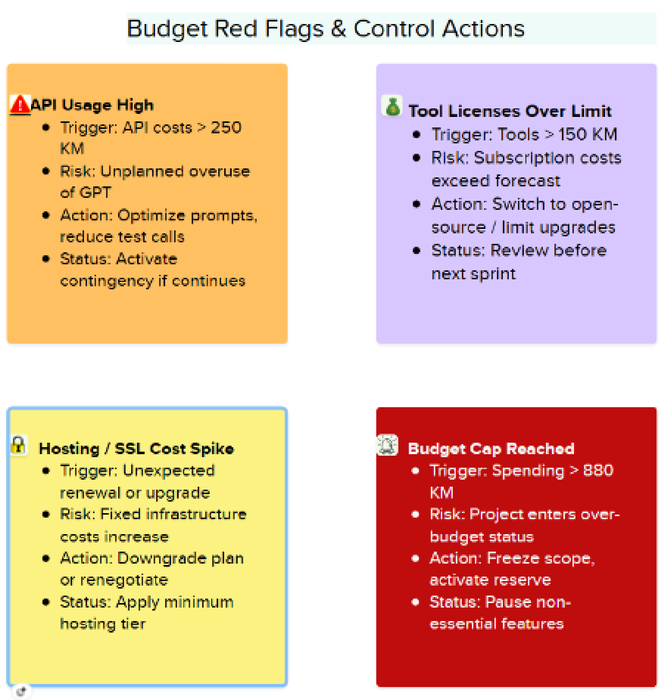

13. Project Development Process

The development of the AI EduAsistent platform follows an Agile methodology, specifically the Scrum framework, allowing for iterative progress, frequent feedback, and the ability to adapt quickly to changes. The process is divided into six main phases, each with clearly defined activities, roles, and outputs. The structure ensures that the product remains aligned with user needs and is delivered efficiently and responsibly.

13.1 Project Phases and Flow

1. **Initiation & Planning**
    - Defined key objectives, functionalities, and timeline.
    - Created a high-level project charter, WBS, and stakeholder map.
    - Tools: Mural (planning), Trello (initial backlog), Google Docs (charter).
2. **Requirements & Design**
    - Collected and structured user requirements (URS).
    - Defined technical architecture (frontend, backend, database).
    - Used MoSCoW prioritization to manage scope.
    - Tools: Visual Studio Code, Miro, draw.io for architecture sketches.
3. **Development**
    - Split into 6 two-week sprints.
    - Frontend, backend, and GPT-3 API integration developed iteratively.
    - Daily standups ensured transparency and progress tracking.
    - Tools: Trello, GitHub, OpenAI Playground, phpMyAdmin.
4. **Testing & Feedback**
    - Conducted internal testing after each sprint.
    - Teachers (pilot group) involved in testing educational content and UX.
    - QA focused on bug fixing, security validation, and form behavior.
    - Tools: Browser Dev Tools, manual test logs, feedback forms.
5. **Deployment & Documentation**
    - Final testing and performance optimization completed.
    - Created user manuals, training materials, and internal documentation.
    - Platform deployed to production-ready hosting.
    - Tools: WordPress (landing page), GitHub, file hosting service.
6. **Closure & Evaluation**
    - Sprint 6 ends with full evaluation of project goals vs OKRs.
    - Final retrospective conducted to reflect on development process.
    - All metrics tracked and reported (KPIs, adoption, API usage).
    - Final report delivered; project officially closed.

13.2 Scrum Roles and Responsibilities

Role	Responsibility
Project Lead	Oversees all phases, approves scope and budget
Frontend Dev	UI/UX development, responsiveness, form validation
Backend Dev	Logic, database connection, GPT-3 integration
QA Tester	Testing functionality, logging bugs, validating fixes
Content Designer	Writes AI literacy modules, checks pedagogical clarity
Mentor/Supervisor	Reviews progress, provides academic and structural feedback

13.3 Tools Used per Phase

Phase	Tools
Planning	Mural, Google Docs, Trello
Requirements	MoSCoW sheets, Miro, draw.io
Development	Visual Studio Code, GitHub, phpMyAdmin, OpenAI API
Testing	Manual test logs, Browser Dev Tools
Deployment	Hosting provider, SSL tools, documentation platforms
Evaluation & Closure	Feedback forms, system logs, metrics dashboard

This structured development process ensured that the platform was continuously evolving while staying aligned with user needs and project objectives. The application of Agile and Scrum practices allowed for quick iterations, effective team communication, and a product that is both technically sound and pedagogically valuable.

14. Scaling and Portfolio Integration

To extend the platform's impact beyond pilot use and ensure institutional adoption, the AI EduAsistent project must scale. This section outlines the changes required for large-scale deployment, including budget implications, new stakeholders, and alignment with broader portfolio objectives.

14.1 Budget Impact of Scaling

Scaling the platform would introduce new cost categories:

- **Infrastructure scaling**: Server capacity, security hardening
- **Multi-team coordination**: Additional developers, product owners, and scrum masters
- **Maintenance of shared AI resources** (e.g., API quotas, fine-tuned models)
- **Training and onboarding**: For new institutional users and stakeholders

The budget would transition from 880 KM (pilot) to a projected **15,000–25,000 KM/year**, depending on user base size, language expansion, and hosting needs.

14.2 New Stakeholders and Roles

Role / Stakeholder	Responsibility
Platform Portfolio Manager	Aligns project OKRs with department-level goals
AI Ethics Board	Ensures ethical compliance at scale
Institutional Deployment Teams	Coordinate implementation in schools
Data Security Lead	Maintains GDPR and national data law compliance
Training Coordinator	Designs and delivers teacher workshops

14.3 Platform-Level OKRs

Objective	Key Results
Promote responsible AI adoption in schools	• Launch AI tools in 25% of primary schools by 2026• Train 500+ teachers via MOOC by year-end• Maintain <3% incident rate in AI tool misuse
Modernize curriculum through AI-powered support	• Implement 3 AI applications in official learning materials• Reach 80% user satisfaction rate across all platforms

14.4 Alignment with AI EduAsistent OKRs

EduAsistent OKR	Connected Platform KR
Enable content generation via GPT	Contributes to adoption of AI in classroom practice
Educate teachers on AI literacy	Supports national teacher training targets
Achieve 50 views on educational pages	Scales into platform-level engagement metrics

15. Critical Project Success Factors

Throughout the lifecycle of AI EduAsistent, several variables influence the quality, usability, and long-term success of the platform. However, three factors stand out as most critical for ensuring that the project delivers meaningful educational value and reaches its core objectives.

15.1 Alignment with Teacher Needs

The platform's main value lies in its ability to assist real teachers with real problems. If the system doesn't reflect their classroom realities,lesson formats, terminology, or educational standards,it will not be adopted. Ongoing feedback loops, pilot testing, and flexible prompt design are essential to stay aligned with their expectations.

Why it matters:
Misalignment results in low usage, wasted development effort, and failure to achieve business or educational impact.

15.2 API Access and Cost Stability

Since the core functionality depends on OpenAI's GPT-3 model, any sudden change in API availability, pricing, or limitations can affect content generation or inflate costs. Continuous monitoring, usage optimization, and having a contingency plan are key.

Why it matters:
If the API becomes too costly or unavailable, the platform loses its core functionality and may no longer be viable.

15.3 Teacher AI Literacy and Training

Even the most powerful tool is ineffective if users don’t understand how to use it. Supporting teachers in learning how to engage with AI,ethically and effectively,is crucial for adoption. This is achieved through educational modules, onboarding materials, and direct support.

Why it matters:
Without AI literacy, teachers may avoid or misuse the system, which can lead to resistance, ethical concerns, or reputational risk.

By closely monitoring these three factors, I ensure that development remains grounded in real user needs, technically sustainable, and aligned with the long-term goal of responsible AI integration in education.